

Customer Shopping Behavior Analysis

From 3,900 purchases to practical growth decisions

How a cleaned transaction dataset becomes a sharper view of value, loyalty, product demand, and customer experience.



The dataset is decision-ready — if behavior is connected to action

01

3,900 purchases

create a broad base for behavioral analysis.

02

18 fields

connect customer context, product choice, offer use, fulfillment, and loyalty signals.

03

Actionable by design

Turn those signals into subscription, loyalty, pricing, positioning, and targeting moves.



TRANSACTION ANATOMY

One transaction record carries the full customer journey

Who

Age, gender, location, and subscription status.

What

Item, category, amount, season, size, and color.

How

Discount use, purchase frequency, review rating, and shipping type.

The cleaning work protects the decisions that follow

01

Load & inspect

Load and inspect the source in Python.

02

Impute

Impute 37 missing review-rating values with the median rating for each product category.

03

Standardize

Standardize column names and engineer age groups plus purchase-frequency days.

04

Remove redundancy

Remove the redundant promo-code field after checking it against discount-applied behavior.

05

Load for analysis

Load the cleaned data into PostgreSQL for structured analysis.

Ten questions turn a transaction table into a business agenda

01

Value

Revenue by gender, age group, shipping type, and subscription status.

03

Products

Top-rated products and top three products in each category.

02

Offers

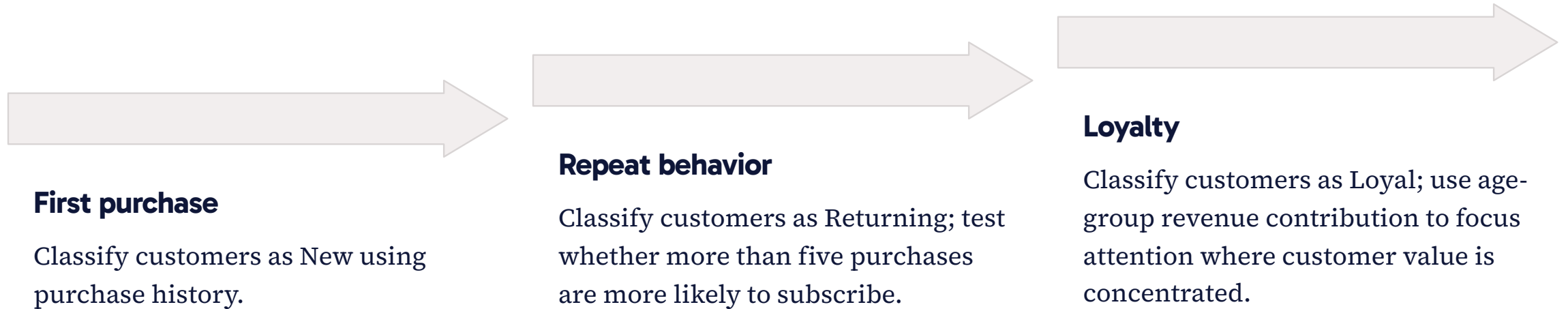
High-spending discount users and discount-dependent products.

04

Loyalty

New, Returning, and Loyal segments plus repeat-buyer subscription likelihood.

Customer value is a journey, not a single segment



Offer performance needs both spend and dependence in view

Profitable promotion

Compare high-spending customers who used discounts with the average purchase amount.

Spend lens

Discount dependence

Identify the five products with the highest share of discounted purchases.

Product lens

Subscription economics

Compare subscribers and non-subscribers on average spend and total revenue.

Customer lens

Product, rating, and delivery signals belong in the same conversation



Ratings

Surface the five products with the highest average review ratings.



Demand

Rank the three most purchased products within each category.



Delivery

Compare average purchase amounts for Standard and Express shipping.

The Power BI dashboard turns analysis into a repeatable operating view

The report closes the loop with an interactive Power BI dashboard that brings the customer, product, offer, loyalty, and fulfillment questions into one visual layer.

PORTFOLIO VIEW

Customer

Move from portfolio view to customer detail.

MERCHANDISING

Product

Move from category to product detail.

SHARED DEFINITIONS

One view

Keep the same definitions across marketing, merchandising, loyalty, and operations.

Five moves connect the analysis to growth

From evidence to action: make the next decision visible, measurable, and targeted.



01 Boost subscriptions

Make exclusive benefits visible to high-value customers.

02 Build loyalty

Reward repeat buyers and move more customers toward the Loyal segment.

03 Rebalance discounts

Protect margin while preserving the sales lift.

04 Position products

Put top-rated and best-selling products into campaigns.

05 Target with evidence

Prioritize high-revenue age groups and Express-shipping users.